Measure of Dispersion

Types of Dispersion

(i) Absolute Measures

(ii) Relative Measure

Absolute Measure is four types —

(i) Range (R)

(ii) Mean Deviation (MD)

(iii) Quartile Deviation (QD)

(iv) Standard Deviation (SD)

Range, $R = H - L$ (Raw + group)

MD is three types –

(i) MD from Mean, $MD(\bar{x})$

(ii) MD " Median, $MD(Median)$

(iii) MD " Mode, $MD(Mode)$

$$MD(\bar{x}) = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n} \quad \text{(Raw)}$$

$$MD(\bar{x}) = \frac{\sum_{i=1}^{n} f_i |x_i - \bar{x}|}{\sum_{i=1}^{n} f_i} \quad \text{(Group)}$$

$$MD(\text{Median}) = \frac{\sum_{i=1}^{n} |x_i - \text{Median}|}{n} \quad (\text{Raw})$$

$$MD(\text{Median}) = \frac{\sum_{i=1}^{n} f_i |x_i - \text{Median}|}{\sum_{i=1}^{n} f_i} \quad (\text{Group})$$

$$MD(\text{Mode}) = \frac{\sum_{i=1}^{n} |x_i - \text{Mode}|}{n} \quad (\text{Raw})$$

$$MD(\text{Mode}) = \frac{\sum_{i=1}^{n} f_i |x_i - \text{Mode}|}{\sum_{i=1}^{n} f_i} \quad (\text{Group})$$

$$QD = \frac{Q_3 - Q_1}{2} \quad (\text{Raw} + \text{group})$$

where, Q_3 = Third Quartile

Q_1 = First "

$$SD = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}} \quad \text{(Raw)}$$

$$SD = \sqrt{\frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}} \quad \text{(Group)}$$

Relative Measure is four types—

(i) Coefficient of Range (CR)

(ii) " " MD (CMD)

(iii) " " QD (CQD)

(iv) " " Variation (CV)

$$CR = \frac{H-L}{H+L} \times 100 \text{ (Raw + group)}$$

CMD is three types—

(i) CMD from mean, CMD ($\bar{x}$)

(ii) CMD from Median, CMD(Median)

(iii) CMD from Mode, CMD (Mode)

$$CMD(\bar{x}) = \frac{MD(\bar{x})}{\bar{x}} \times 100 \text{ (Raw + group)}$$

$$CMD(Mode) = \frac{MD(Mode)}{Mode} \times 100 \text{ (Raw + group)}$$

$$CMD(Median) = \frac{MD(Median)}{Median} \times 100$$

(Raw + group)

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

(Raw + group)

$$CV = \frac{SD}{\bar{x}} \times 100$$ (Raw + group)